

## ORIGINAL CONTRIBUTION

# Thrombolysis-to-Puncture Time Determines the Differential Effect of Tenecteplase Versus Alteplase in Large Vessel Occlusion

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**BACKGROUND:** Tenecteplase improves reperfusion and functional outcomes compared with alteplase in patients with large vessel occlusion. We assessed whether this superiority varies with thrombolysis-to-puncture time (TTP).

**METHODS:** This retrospective analysis of a prospective multicenter cohort included patients with large vessel occlusion who received bridging therapy with either tenecteplase or alteplase between January 2022 and September 2025. Patients were stratified by TTP (<60 versus ≥60 minutes). We subsequently assessed the association of TTP with efficacy and safety outcomes between the alteplase and tenecteplase groups using multivariable logistic regression adjusted for age, baseline National Institutes of Health Stroke Scale score, and time from stroke onset to arterial puncture. Functional independence was defined as a modified Rankin Scale score of 0 to 2 at 3 months.

**RESULTS:** Of 1106 patients who received bridging therapy, 1003 were included in the primary analysis (tenecteplase, 488; alteplase, 515). The median age was 68 (interquartile range, 58–75) years in both groups, with males comprising 64.3% and 66.0%, respectively. Tenecteplase was associated with superior 3-month functional independence compared with alteplase (53.6% versus 48.1%; adjusted odds ratio, 1.37 [95% CI, 1.03–1.82]). This benefit was concentrated in patients with a TTP <60 minutes, where tenecteplase yielded higher rates of both early recanalization (19.0% versus 9.1%; adjusted odds ratio, 2.36 [95% CI, 1.32–4.23];  $P_{\text{interaction}}=0.047$ ) and functional independence (61.0% versus 49.0%; adjusted odds ratio, 1.77 [95% CI, 1.13–2.77];  $P_{\text{interaction}}=0.111$ ). No between-agent differences were significant with TTP ≥60 minutes. Safety outcomes were comparable, but each 30-minute TTP increase independently elevated hemorrhagic risks for both agents.

**CONCLUSIONS:** The recanalization superiority of tenecteplase over alteplase is time-dependent, evident only within a TTP <60 minutes. Although the translation of this advantage into functional outcome was not statistically modified by time, optimizing workflow to achieve this rapid window maximizes the potential benefit of tenecteplase, which should be prioritized in capable settings.

**GRAPHIC ABSTRACT:** A [graphic abstract](#) is available for this article.

**Key Words:** functional status ■ half-life ■ ischemic stroke ■ tenecteplase ■ tissue plasminogen activator

Endovascular thrombectomy (EVT) is the standard treatment for acute ischemic stroke caused by large vessel occlusion (LVO).<sup>1</sup> For eligible patients presenting within 4.5 hours, guidelines recommend intravenous thrombolysis (IVT) as bridging therapy before EVT.<sup>2</sup> This

approach, supported by multiple randomized controlled trials, exerts the benefit of IVT by inducing early recanalization, improving distal microcirculatory flow, and softening the thrombus to facilitate subsequent mechanical removal.<sup>3–6</sup>

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## Nonstandard Abbreviations and Acronyms

|             |                                     |
|-------------|-------------------------------------|
| <b>aOR</b>  | adjusted odds ratio                 |
| <b>EVT</b>  | endovascular thrombectomy           |
| <b>IVT</b>  | intravenous thrombolysis            |
| <b>LVO</b>  | large vessel occlusion              |
| <b>mRS</b>  | modified Rankin Scale               |
| <b>PH-2</b> | parenchymal hematoma type 2         |
| <b>slCH</b> | symptomatic intracranial hemorrhage |
| <b>TTP</b>  | thrombolysis-to-puncture time       |

Bridging therapy outcomes hinge on workflow efficiency, with thrombolysis-to-puncture time (TTP) as a key modifiable factor.<sup>7</sup> For alteplase, every 30-minute delay in TTP reduces functional independence, lowers reperfusion rates, and increases symptomatic intracranial hemorrhage (slCH) risk,<sup>8,9</sup> making TTP a critical quality metric in stroke care. Tenecteplase offers a distinct pharmacological profile versus alteplase, including greater fibrin specificity and a longer half-life.<sup>10</sup> While proven noninferior overall,<sup>11–14</sup> its potential superiority emerges in LVO bridging therapy, where it is associated with higher rates of early recanalization.<sup>15</sup> Although this benefit has been reported to be present regardless of the time from administration to reperfusion assessment across a broad range of delays, its magnitude appears most pronounced with shorter TTP.<sup>16,17</sup> A critical unanswered question, therefore, is whether the early efficacy advantage of tenecteplase is itself time-dependent within the clinically decisive window preceding thrombectomy. The pharmacological profiles of the 2 agents suggest fundamentally different time-efficacy relationships: tenecteplase's single-bolus, front-loaded action may decay with procedural delays,<sup>18</sup> whereas alteplase's continuous infusion may yield a more cumulative effect.<sup>19,20</sup>

Therefore, this study aimed to compare the differential impact of TTP on the efficacy and safety of tenecteplase versus alteplase in patients with LVO undergoing bridging therapy. We hypothesized that the superiority of tenecteplase over alteplase is time-dependent, being most pronounced with shorter TTP intervals.

## METHODS

Data collected for the study may be made available from the corresponding author to others upon reasonable request.

### Study Design and Population

We conducted a retrospective analysis of a prospective, multicenter cohort that included consecutive patients receiving IVT and EVT for acute ischemic stroke across 7 comprehensive stroke centers in China (the list of centers is detailed in

Table S1). This registry aimed to characterize the real-world clinical practice of IVT and EVT in the Chinese population. All participating centers maintained 24/7 thrombectomy services, and each certified site was required to perform a minimum of 100 IVT and 50 EVT procedures annually. Following the 2023 Chinese stroke guideline update that accorded tenecteplase and alteplase equal recommendation, the selection between these agents in clinical practice was guided by evolving institutional protocols, with stroke centers gradually transitioning from alteplase to tenecteplase as their primary thrombolytic agent over time. The study protocol was approved by the institutional ethics committees of all participating centers ([2021]215). Written informed consent was obtained from all patients or their legally authorized representatives, and the study was conducted in accordance with the ethical principles of the Declaration of Helsinki. This study was reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.

Patients enrolled in the study between January 2022 and September 2025 were included in the analysis if they met the following criteria: (1) baseline computed tomography angiography confirming an LVO in the internal carotid artery, the first segment of the middle cerebral artery/the second segment of the middle cerebral artery or basilar artery, eligible for bridging therapy; (2) were directly admitted to an EVT-capable center (mothership model); (3) received IVT with either standard-dose alteplase (0.9 mg/kg) or tenecteplase (0.25 mg/kg) before EVT; and (4) had a reliably documented interval from thrombolysis to groin puncture.

The exclusion criteria were as follows: (1) a time from thrombolysis to arterial puncture exceeding 180 minutes; (2) cases with inadequate image quality or substantial missing clinical data; (3) patients transferred to another hospital without available follow-up imaging data; (4) presence of a concomitant neurological disorder that could confound neurological assessment (eg, significant prestroke disability [modified Rankin Scale (mRS) score > 2], severe dementia, or primary intracranial hemorrhage); and (5) a severe systemic comorbidity known to substantially influence functional outcome and survival (eg, advanced malignancy or end-stage renal disease).

### Data Collection

Demographic, clinical, imaging, and procedural data were prospectively collected using an electronic data capture system (<https://5gmsu.com/login>). An independent neuroradiologist, blinded to clinical details, assessed all imaging for occlusion site, intracranial hemorrhage (ICH), and preprocedural and postprocedural modified Thrombolysis in Cerebral Infarction scores. Functional outcome was assessed at 3 months using the mRS score via telephone interview or structured in-person follow-up.

### Definition of Outcome Variables

The imaging outcome was the reperfusion status, graded using the modified Thrombolysis in Cerebral Infarction scale score on both the initial and final digital subtraction angiography series. Successful recanalization was defined as a modified Thrombolysis in Cerebral Infarction score grade of 2b to 3, indicating reperfusion of over 50% of the affected territory, or the

absence of a retrievable thrombus on the initial digital subtraction angiography run.

Clinical outcomes were assessed using the National Institutes of Health Stroke Scale score at 24 hours and the mRS score at 3 months. Early neurological improvement was defined as a reduction in National Institutes of Health Stroke Scale score by  $\geq 8$  points or achieving a score of 0 to 1 at 24 hours. Functional independence was defined as an mRS score of 0 to 2 at 90 days. The 3-month mRS scores were determined by certified neurologists via structured telephone interviews or during routine outpatient visits. The assessors of these clinical outcomes were blinded to the patient's care pathway and treatment details.

Safety outcomes comprised parenchymal hematoma type 2 (PH-2) per ECASS II criteria,<sup>21</sup> sICH per SITS-MOST criteria,<sup>22</sup> and all-cause mortality (defined as an mRS score of 6).

## Statistical Analysis

Continuous variables are presented as median (interquartile range) and were compared using the Mann-Whitney *U* test. Categorical variables are presented as counts (percentages) and were compared using the  $\chi^2$  or Fisher exact test, as appropriate. Missing data were handled using listwise deletion in the relevant analyses.

To evaluate the association between TTP and clinical outcomes, we used 2 complementary analytical approaches. Patients were first stratified into 2 groups (TTP  $< 60$  versus  $\geq 60$  minutes), with between-group differences in efficacy and safety outcomes assessed using both unadjusted and adjusted models. Second, we modeled the continuous relationship between TTP and efficacy outcomes (early recanalization and 3-month functional independence) using multivariable-adjusted restricted cubic splines (with 3 knots) to capture potential nonlinear patterns, with separate curves fitted for each thrombolytic agent. In contrast, for safety outcomes (sICH and PH-2), we assumed a linear association and used multivariable logistic regression to estimate the adjusted odds ratio (aOR) per 30-minute increase in TTP. To assess whether the treatment effect was modified by TTP, we tested the interaction between thrombolytic type and TTP, dichotomized as  $< 60$  versus  $\geq 60$  minutes, in the corresponding regression models for early recanalization and 3-month functional independence.

All adjusted models included the following prespecified covariates: age, baseline National Institutes of Health Stroke Scale score, and time from stroke symptom onset to arterial puncture. All statistical analyses were performed using R software (version 4.2.2). A 2-sided  $P < 0.05$  was considered statistically significant.

## RESULTS

A total of 3510 consecutive patients with acute ischemic stroke received IVT. Of this cohort, 1106 patients underwent bridging therapy, which included 576 treated with 0.9-mg/kg alteplase and 530 with 0.25-mg/kg tenecteplase. After applying exclusion criteria (missing digital subtraction angiography, computed tomography angiography data, or undocumented TTP time), the alteplase and tenecteplase groups comprised 515 and

488 patients, respectively, eligible for further analysis. With the subsequent exclusion of 60 and 55 patients lost to 3-month follow-up, the final mRS analysis included 455 patients in the alteplase group and 433 in the tenecteplase group (Figure S1).

The baseline characteristics of the 1003 patients stratified by TTP were summarized in Table 1. The 2 groups were well-balanced in terms of demographic and clinical characteristics, including age, sex, baseline National Institutes of Health Stroke Scale score, medical history, site of arterial occlusion, cause of stroke, and Alberta Stroke Program Early Computed Tomography Score (all  $P > 0.05$ ). Regarding workflow times, the intervals from stroke onset to thrombolysis and from puncture to recanalization were similar between groups. In contrast, the time from stroke onset to puncture was significantly longer in the  $\geq 60$ -minute group (median, 275 versus 210 minutes;  $P < 0.001$ ), consistent with the defining group difference in the thrombolysis-to-puncture interval (median, 90 versus 44 minutes;  $P < 0.001$ ). A comparison of baseline characteristics between the tenecteplase and alteplase groups is provided in Table S2.

In the overall cohort, tenecteplase was associated with a higher rate of functional independence (mRS score, 0–2) at 3 months compared with alteplase (53.6% [232/433] versus 48.1% [219/455]; aOR, 1.37 [95% CI, 1.03–1.82];  $P = 0.031$ ). The comparative efficacy stratified by TTP is detailed in Table 2. For the outcome of early recanalization, the benefit of tenecteplase was entirely concentrated in the  $< 60$ -minute stratum (19.0% versus 9.1%; aOR, 2.36 [95% CI, 1.32–4.23]), with no significant difference in the  $\geq 60$ -minute stratum (20.8% versus 19.3%; aOR, 1.16 [95% CI, 0.77–1.76]). A significant treatment-by-time interaction was confirmed ( $P_{\text{interaction}} = 0.047$ ). For the outcome of 3-month functional independence, similarly favored tenecteplase in the  $< 60$ -minute stratum (61.0% versus 49.0%; aOR, 1.77 [95% CI, 1.13–2.77]), with no significant difference in the  $\geq 60$ -minute stratum (48.0% versus 47.5%; aOR, 1.12 [95% CI, 0.77–1.62]). However, the interaction between treatment effect and TTP for this clinical end point was not statistically significant ( $P_{\text{interaction}} = 0.111$ ). Other outcomes, including post-thrombectomy recanalization, early neurological improvement, and the distribution of 3-month mRS scores, were comparable between treatment groups in both TTP strata. The distribution of the mRS scores at 3 months is shown in Figure 1. Detailed rates of early recanalization and functional independence, stratified by TTP interval for each agent, are provided in Tables S4 and S5.

Restricted cubic spline analyses revealed distinct patterns in the association between TTP and early recanalization for the 2 thrombolytic agents (Figure 2). For alteplase (Figure 2A), a nonlinear relationship was

**Table 1. Baseline Characteristics**

|                                             | <60-min<br>thrombolysis-<br>to-puncture<br>time (n=424) | ≥60-min<br>thrombolysis-<br>to-puncture<br>time (n=579) | P value |
|---------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|---------|
| Age, y; median (IQR)                        | 68 (58–75)                                              | 68 (58–75)                                              | 0.489   |
| Male, n (%)                                 | 282 (66.5%)                                             | 372 (64.2%)                                             | 0.458   |
| NIHSS score, median (IQR)                   | 16 (12–19)                                              | 16 (12–20)                                              | 0.574   |
| Medical history, n (%)                      |                                                         |                                                         |         |
| Hypertension                                | 271 (63.9%)                                             | 390 (67.4%)                                             | 0.256   |
| Hyperlipidemia                              | 118 (27.8%)                                             | 140 (24.2%)                                             | 0.191   |
| Diabetes                                    | 93 (21.9%)                                              | 147 (25.4%)                                             | 0.205   |
| Coronary heart disease                      | 113 (26.7%)                                             | 150 (25.9%)                                             | 0.791   |
| Atrial fibrillation                         | 103 (24.3%)                                             | 136 (23.5%)                                             | 0.768   |
| Previous stroke                             | 104 (24.5%)                                             | 140 (24.2%)                                             | 0.899   |
| Antiplatelets                               | 98 (23.1%)                                              | 136 (23.5%)                                             | 0.890   |
| Current smoking                             | 137 (32.3%)                                             | 172 (29.7%)                                             | 0.377   |
| Site of occlusion, n (%)                    |                                                         |                                                         | 0.301   |
| ICA                                         | 123 (29.0%)                                             | 161 (27.8%)                                             |         |
| M1                                          | 162 (38.2%)                                             | 198 (34.2%)                                             |         |
| M2                                          | 88 (20.8%)                                              | 148 (25.6%)                                             |         |
| BA                                          | 51 (12.0%)                                              | 72 (12.4%)                                              |         |
| Cause of stroke, n (%)                      |                                                         |                                                         | 0.474   |
| Large artery atherosclerosis (extracranial) | 60 (14.2%)                                              | 87 (15.0%)                                              |         |
| Large artery atherosclerosis (intracranial) | 183 (43.2%)                                             | 225 (38.9%)                                             |         |
| Cardioembolic occlusion                     | 147 (34.7%)                                             | 208 (35.9%)                                             |         |
| Unknown                                     | 24 (5.7%)                                               | 47 (8.1%)                                               |         |
| Others                                      | 10 (2.4%)                                               | 12 (2.1%)                                               |         |
| ASPECTS, median (IQR)                       | 8 (7, 9)                                                | 8 (7, 9)                                                | 0.579   |
| Workflow times, min; median (IQR)           |                                                         |                                                         |         |
| Stroke symptom onset to thrombolysis*       | 170 (123–240)                                           | 170 (112–238)                                           | 0.328   |
| Stroke symptom onset to puncture†           | 210 (165–284)                                           | 275 (210–345)                                           | <0.001  |
| Thrombolysis to puncture                    | 44 (34–50)                                              | 90 (73–125)                                             | <0.001  |
| Puncture to recanalization‡                 | 35 (25–51)                                              | 33 (23–52)                                              | 0.699   |

ASPECTS indicates Alberta Stroke Program Early Computed Tomography Score; BA, basilar artery; ICA, internal carotid artery; IQR, interquartile range; IVT, intravenous thrombolysis; M1, the first segment of the middle cerebral artery; M2, the second segment of the middle cerebral artery; and NIHSS, National Institutes of Health Stroke Scale.

Number of missing values:

\*2, †2, and ‡64.

identified ( $P_{\text{nonlinear}}=0.002$ ), with a threshold effect at 65 minutes ( $P$  value for likelihood ratio test=0.001). Each 1-minute increase in TTP below this threshold was associated with an aOR of 1.05 (95% CI, 1.01–1.09;  $P=0.010$ ) for early recanalization, whereas no significant association was observed beyond 65 minutes (aOR, 0.99, [95% CI, 0.98–1.00];  $P=0.196$ ). In contrast, the probability remained stable over time for tenecteplase

(aOR per 1-minute increase, 1.00 [95% CI, 0.99–1.01]; Figure 2B).

A similarly divergent relationship was observed for the outcome of functional independence (Figure 3). For alteplase (Figure 3A), a nonlinear association was detected ( $P_{\text{nonlinear}}=0.006$ ), with an inflection point near 60 minutes. Each 30-minute increase in TTP within the first 60 minutes was associated with an aOR of 1.03 (95% CI, 1.01–1.06;  $P=0.012$ ) for functional independence, while each 30-minute increase beyond 60 minutes was associated with an aOR of 0.71 (95% CI, 0.55–0.94;  $P=0.015$ ). For tenecteplase (Figure 3B), a linear association was observed ( $P_{\text{overall}}=0.010$ ), with each 30-minute increase in TTP corresponding to an aOR of 0.78 (95% CI, 0.65–0.93;  $P=0.005$ ) for functional independence.

Safety outcomes are shown in Table 3. The incidence of PH-2, sICH, and 3-month mortality was similar between the tenecteplase and alteplase groups in both the <60- and ≥60-minute strata. However, compared with the <60-minute stratum, the ≥60-minute stratum showed a higher incidence of sICH and PH-2 in both treatment groups. The TTP was significantly associated with an increased risk of hemorrhagic complications for both thrombolytic agents. Each 30-minute increase in time was associated with higher odds of sICH, with an aOR of 1.51 (95% CI, 1.12–2.03;  $P=0.006$ ) for alteplase and 1.35 (95% CI, 1.01–1.79;  $P=0.043$ ) for tenecteplase (Figure 4A). Similarly, the probability of PH-2 increased with longer time intervals for both agents, with aORs of 1.24 (95% CI, 1.04–1.49;  $P=0.018$ ) per 30-minute increase for alteplase and 1.38 (95% CI, 1.10–1.72;  $P=0.005$ ) for tenecteplase (Figure 4B).

## DISCUSSION

This multicenter cohort study demonstrated that the comparative efficacy of tenecteplase versus alteplase in patients with LVO undergoing bridging therapy is critically dependent on the TTP. We reported 3 principal findings: first, tenecteplase was associated with superior functional independence at 3 months in the overall cohort; second, this advantage was concentrated in the subgroup with a TTP of <60 minutes, wherein tenecteplase yielded significantly higher rates of early recanalization and a strong trend toward better functional independence; and third, the risks of PH-2 and sICH increased significantly with longer TTP for both thrombolytic agents.

Current evidence from large-scale randomized controlled trials supported the noninferiority of tenecteplase relative to alteplase in acute ischemic stroke, while its superior efficacy was restricted to specific patient subgroups.<sup>11–14</sup> Specifically, tenecteplase has been associated with improved 3-month functional outcomes in



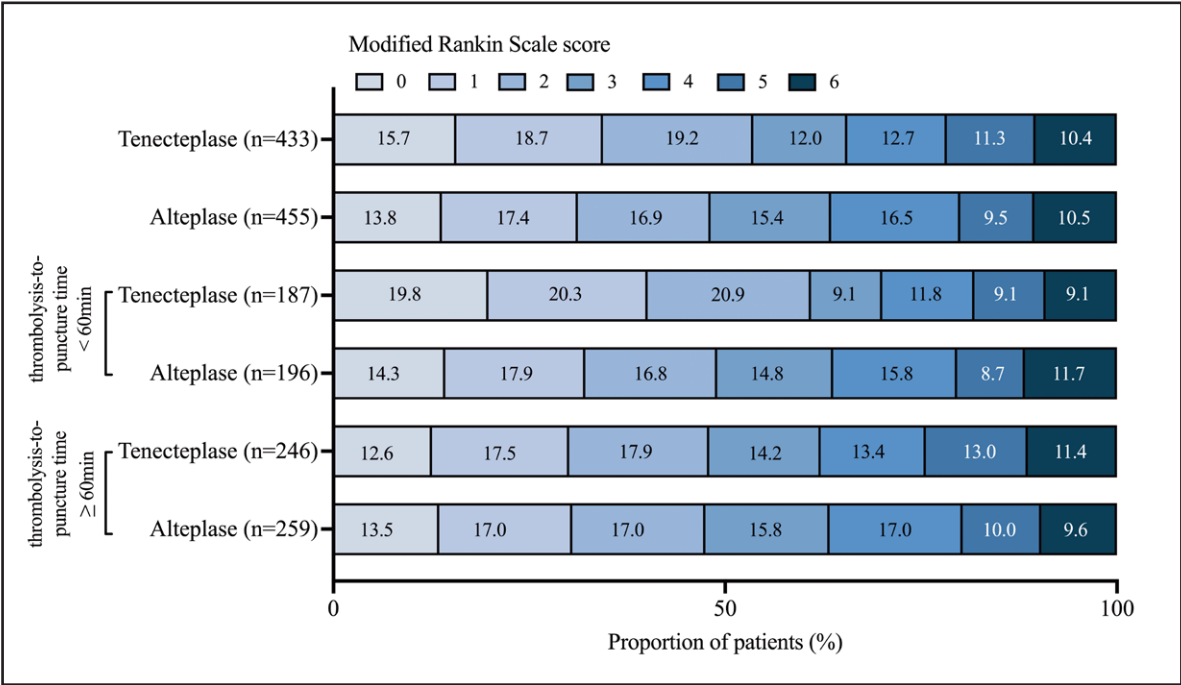
**Table 2. Efficacy Outcomes of Tenecteplase and Alteplase, Stratified by Thrombolysis-to-Puncture Time (<60 and ≥60 min)**

|                                       | <60-min thrombolysis-to-puncture time |                      |                        |                      | ≥60-min thrombolysis-to-puncture time |                      |                        |                      |
|---------------------------------------|---------------------------------------|----------------------|------------------------|----------------------|---------------------------------------|----------------------|------------------------|----------------------|
|                                       | Alteplase (n=219)                     | Tenecteplase (n=205) | Unadjusted OR (95% CI) | Adjusted OR (95% CI) | Alteplase (n=296)                     | Tenecteplase (n=283) | Unadjusted OR (95% CI) | Adjusted OR (95% CI) |
| Early recanalization before MT, n (%) | 20/219 (9.1%)                         | 39/205 (19.0%)       | 2.34 (1.31–4.16)       | 2.36 (1.32–4.23)     | 57/296 (19.3%)                        | 59/283 (20.8%)       | 1.10 (0.74–1.66)       | 1.16 (0.77–1.76)     |
| Post-MT recanalization, n (%)         | 175/205 (85.4%)                       | 160/182 (87.9%)      | 1.25 (0.69–2.25)       | 1.27 (0.70–2.31)     | 214/254 (84.3%)                       | 205/237 (86.5%)      | 1.20 (0.72–1.98)       | 1.21 (0.73–2.01)     |
| Early neurological improvement, n (%) | 60/219 (27.4%)                        | 49/205 (23.9%)       | 0.83 (0.54–1.29)       | 0.88 (0.56–1.38)     | 60/296 (20.3%)                        | 66/283 (23.3%)       | 1.20 (0.81–1.78)       | 1.14 (0.76–1.71)     |
| mRS score, 0–1 at 90 d, n (%)*        | 63/196 (32.1%)                        | 75/187 (40.1%)       | 1.41 (0.93–2.15)       | 1.43 (0.92–2.24)     | 79/259 (30.5%)                        | 74/246 (30.1%)       | 0.98 (0.67–1.43)       | 1.05 (0.70–1.56)     |
| mRS score, 0–2 at 90 d, n (%)*        | 96/196 (49.0%)                        | 114/187 (61.0%)      | 1.63 (1.08–2.44)       | 1.77 (1.13–2.77)     | 123/259 (47.5%)                       | 118/246 (48.0%)      | 1.02 (0.72–1.45)       | 1.12 (0.77–1.62)     |
| mRS score at 90 d, median (IQR)*      | 3 (1–4)                               | 2 (1–4)              | 0.70 (0.49–1.00)       | 0.67 (0.47–0.97)     | 3 (1–4)                               | 3 (1–4)              | 1.07 (0.79–1.46)       | 0.99 (0.72–1.34)     |

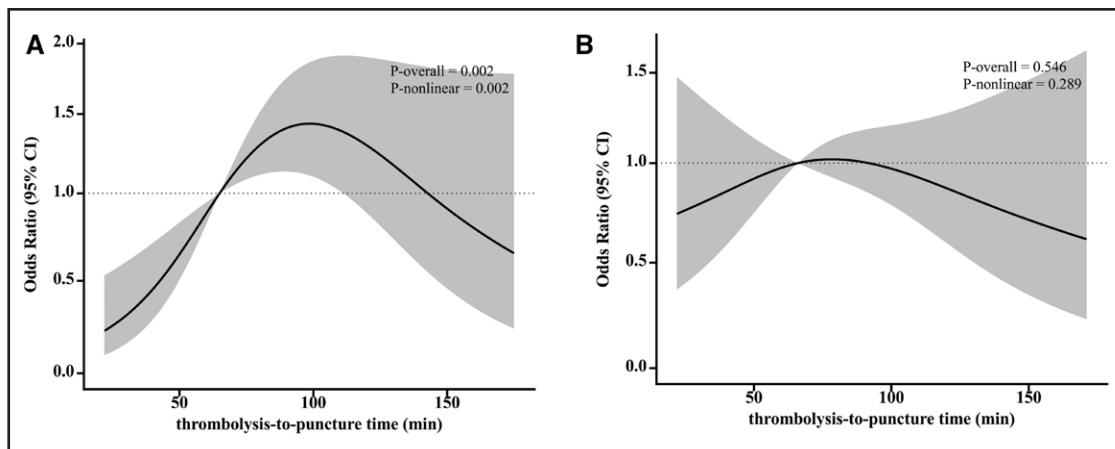
IQR indicates interquartile range; MT, mechanical thrombectomy; and OR, odds ratio.  
\*End point not available in 55 and 60 patients in the tenecteplase and alteplase groups, respectively.

patients with larger ischemic cores and higher early reperfusion rates in those with larger thrombi.<sup>23,24</sup> Our findings defined a critical <60-minute TTP window for tenecteplase’s superiority over alteplase in early recanalization and functional outcomes. This narrow therapeutic window explained the disparities across previous research, wherein tenecteplase proved superior to alteplase under efficient workflows but showed comparable outcomes when delays occurred.<sup>15,25,26</sup> The significant treatment-by-time interaction for early recanalization ( $P_{\text{interaction}}=0.035$ ) confirmed the time-dependent superiority of tenecteplase over alteplase. In contrast, the absence of this interaction for functional independence ( $P_{\text{interaction}}=0.320$ ) likely reflected the substantial influence

of subsequent factors, particularly the ultimate success of thrombectomy and individual collateral status, which modulate long-term outcomes and, thereby, dilute the specific contribution of the initial thrombolytic agent.<sup>27–29</sup> This interpretation is directly supported by our formal mediation analysis, which indicated that early recanalization explained a significant but partial ( $\approx 14\%$ ) proportion of tenecteplase’s total treatment effect (Table S3). The exploratory findings suggested a potential advantage of tenecteplase over alteplase in settings where a short TTP can be achieved. An ongoing clinical trial (Unique identifier: NCT06658197) will further evaluate the efficacy of tenecteplase in rapid bridging therapy before mechanical thrombectomy for large vessel occlusion.



**Figure 1. Modified Rankin Scale scores at 3 months.**



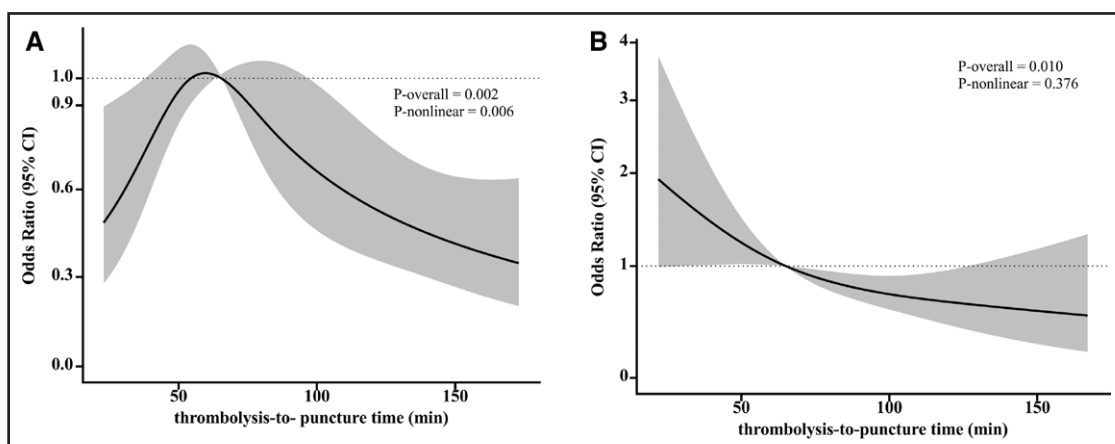
**Figure 2. Association of thrombolysis-to-puncture time with pre-mechanical thrombectomy recanalization in a restricted cubic spline model.**

**A**, Alteplase group. **B**, Tenecteplase group. aORs: solid line. 95% CI: shadows.

The superior efficacy of tenecteplase within shorter TTP windows was consistent with its distinct pharmacological properties.<sup>10,18</sup> Its single-bolus administration and sustained fibrinolytic activity enable rapid accumulation of a high drug concentration at the thrombus site, resulting in a potent, front-loaded effect that promotes early recanalization.<sup>30,31</sup> However, our observation of a flat time-reperfusion relationship differed from the rising association (aOR, 1.08) reported by Yogen-drakumar et al. This divergence may be explained by 2 factors: potential subtle pharmacodynamic differences between the 2 manufacturer formulations. Meanwhile, distinct cohort profiles. Our mothership cohort had a high prevalence of large artery atherosclerosis and internal carotid artery occlusions (thrombi less susceptible to lysis), whereas their study included many drip-and-ship patients enriched in cardioembolic middle cerebral artery occlusions (typically more fibrin-rich and amenable to delayed lysis).<sup>32,33</sup> Consequently, while the rate of early recanalization remained stable over time, the probability of functional independence declined

linearly with prolonged TTP. This decline reflected the progressive and irreversible loss of salvageable penumbra with each minute of delay.<sup>34</sup> Thus, every minute of prolonged TTP undermined the clinical efficacy of tenecteplase.

In contrast, alteplase exhibited a distinct nonlinear relationship with TTP. Although previous studies consistently reported worsening outcomes with prolonged TTP,<sup>3,9,35</sup> our analysis revealed a biphasic pattern: functional independence initially improved as TTP extended to 1 hour and then progressively deteriorated thereafter. This pattern corresponds with the pharmacodynamic profile of alteplase. Its peak fibrinolytic effect, reflected by D-dimer levels, occurs approximately 1 hour after infusion initiation, while its short half-life (5 minutes) leads to rapid clearance upon infusion completion.<sup>19,20,36</sup> This profile supports that alteplase exerts a cumulative lytic effect that develops during the infusion but cannot be sustained after it ends. The initial improvement in functional outcomes likely reflects this accumulating therapeutic effect, whereas the



**Figure 3. Association of thrombolysis-to-puncture time with functional independence in a restricted cubic spline model.**

**A**, Alteplase group. **B**, Tenecteplase group. aORs: solid line. 95% CI: shadows.

**Table 3. Safety Outcomes of Tenecteplase and Alteplase, Stratified by Thrombolysis-to-Puncture Time (<60 and ≥60 min)**

|                           | <60-min thrombolysis-to-puncture time |                      |                        |                      | ≥60-min thrombolysis-to-puncture time |                      |                        |                      |
|---------------------------|---------------------------------------|----------------------|------------------------|----------------------|---------------------------------------|----------------------|------------------------|----------------------|
|                           | Alteplase (n=219)                     | Tenecteplase (n=205) | Unadjusted OR (95% CI) | Adjusted OR (95% CI) | Alteplase (n=296)                     | Tenecteplase (n=283) | Unadjusted OR (95% CI) | Adjusted OR (95% CI) |
| PH-2, n (%)               | 20/219 (9.1%)                         | 18/205 (8.8%)        | 0.96 (0.49–1.87)       | 0.95 (0.48–1.86)     | 44/296 (14.9%)                        | 46/283 (16.3%)       | 1.11 (0.71–1.74)       | 1.05 (0.67–1.67)     |
| slCH, n (%)               | 8/219 (3.7%)                          | 9/205 (4.4%)         | 1.21 (0.46–3.20)       | 1.29 (0.47–3.54)     | 19/296 (6.4%)                         | 21/283 (7.4%)        | 1.17 (0.61–2.22)       | 1.14 (0.60–2.17)     |
| Death within 90 d, n (%)* | 23/196 (11.7%)                        | 17/187 (9.1%)        | 0.75 (0.39–1.46)       | 0.75 (0.36–1.54)     | 25/259 (9.7%)                         | 28/246 (11.4%)       | 1.20 (0.68–2.13)       | 1.12 (0.62–2.04)     |

OR indicates odds ratio; PH-2, parenchymal hematoma type 2; and slCH, symptomatic intracranial hemorrhage.  
\*End point not available in 55 and 60 patients in the tenecteplase and alteplase groups, respectively.

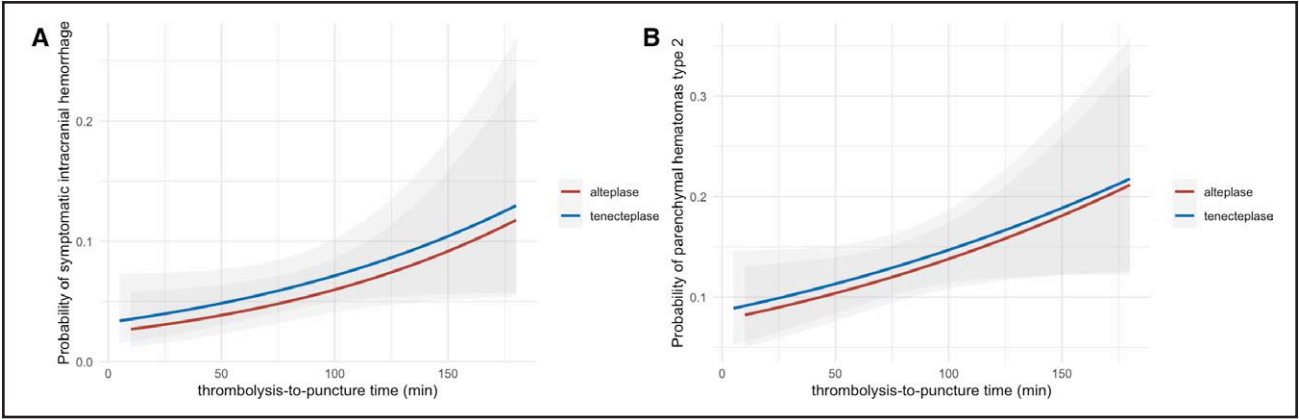
subsequent decline indicates that the benefits of early recanalization are eventually outweighed by the detrimental impact of prolonged ischemia.<sup>37,38</sup> These findings suggest that although alteplase may have a wider initiation window, its therapeutic benefit is ultimately compromised by procedural delays. Consequently, an IVT-to-puncture time exceeding 1 hour represents a pseudobridging scenario, an ineffective waiting period that provides no additional benefit for either thrombolytic agent. Therefore, maximizing workflow efficiency remains imperative for optimizing outcomes with both agents.

The differential time-efficacy relationships identified in our study explain the seemingly contradictory results of recent thrombolysis-omission trials, in which direct thrombectomy was noninferior to alteplase-bridging, whereas tenecteplase-bridging was superior to direct thrombectomy.<sup>39,40</sup> In the trial demonstrating the noninferiority of direct thrombectomy, the short randomization-to-puncture intervals (36 minutes) were insufficient for alteplase to attain its peak lytic effect, thereby limiting its additional benefit. Conversely, similarly brief TTP intervals (16 minutes) in the tenecteplase trial were sufficient to enable its rapid, front-loaded action, thus establishing its superiority. This distinction reaffirms that the therapeutic window for bridging therapy is agent-specific: tenecteplase capitalizes on ultrarapid workflows, whereas

alteplase requires sufficient infusion time to achieve its cumulative effect.

Our analysis of safety outcomes provides further evidence for the association between TTP and hemorrhagic risk in bridging therapy. Although the incidence of slCH and PH-2 did not differ significantly between tenecteplase and alteplase within any TTP stratum, each 30-minute delay in TTP was associated with a significant increase in the risk of both hemorrhagic complications for each agent. While this relationship had been previously established for alteplase, it had not been well characterized for tenecteplase.<sup>8</sup> This consistent observation establishes TTP as an independent risk factor for hemorrhagic transformation, independent of thrombolytic selection, thereby supporting the necessity of optimizing the bridging therapy workflow.

Our study has several limitations. First, as a real-world, observational analysis, it is subject to potential selection bias and unmeasured confounding factors, despite adjustments for key covariates in the statistical models. Second, while adequately powered for primary outcomes, the study had modest statistical power to detect more subtle interaction effects. Third, thrombolysis-to-puncture intervals, though prospectively documented, remain subject to intercenter heterogeneity in recording practices. Finally, the generalizability of these findings requires verification in broader populations and health care settings.



**Figure 4. Individual probabilities according to thrombolysis-to-puncture time.**  
**A,** For symptomatic intracerebral hemorrhage. **B,** For parenchymal hematoma type 2. aORs: solid line. 95% CI: shadows.

## CONCLUSIONS

This study demonstrates that the comparative efficacy of tenecteplase versus alteplase is highly time-sensitive. The superior early recanalization and functional independence associated with tenecteplase were confined to a TTP of <60 minutes, an advantage that was no longer observed with longer delays. Prolonged TTP additionally elevated hemorrhagic risk for both agents. Therefore, optimizing workflow efficiency is paramount, and tenecteplase should be prioritized in clinical settings capable of consistently achieving rapid treatment pathways.

## ARTICLE INFORMATION

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### Author Contributions

Drs Wang, Liu, Ji, and Guo conceived and designed the study. Drs Che, Han, Liu, and Wang contributed to the acquisition of data. Drs Wang, Yang, Wu, Jiao, Li, and Zhang contributed to statistical analyses and interpretation of data. Drs Wang, Yang, and Guo drafted and revised the manuscript for content. All authors have read and approved the manuscript.

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### Disclosures

None.

### Supplemental Material

Tables S1–S5  
Figure S1

## REFERENCES

1. Powers WJ, Rabinstein AA, Ackerson T, Adeoye OM, Bambakidis NC, Becker K, Biller J, Brown M, Demaerschalk BM, Hoh B, et al. Guidelines for the early management of patients with acute ischemic stroke: 2019 update to the 2018 guidelines for the early management of acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*. 2019;50:e344–e418. doi: 10.1161/STR.0000000000000211
2. Berge E, Whiteley W, Audebert H, Marchis GMD, Fonseca AC, Padiglioni C, Ossa NP, Strbian D, Tsvigoulis G, Turc G. European Stroke Organisation (ESO) guidelines on intravenous thrombolysis for acute ischaemic stroke. *Eur Stroke J*. 2021;6:l–LXII. doi: 10.1177/2396987321989865
3. Suzuki K, Matsumaru Y, Takeuchi M, Morimoto M, Kanazawa R, Takayama Y, Kamiya Y, Shigeta K, Okubo S, Hayakawa M, et al; SKIP Study Investigators. Effect of mechanical thrombectomy without vs with intravenous thrombolysis on functional outcome among patients with acute ischemic stroke: the SKIP randomized clinical trial. *JAMA*. 2021;325:244–253. doi: 10.1001/jama.2020.23522
4. LeCouffe NE, Kappelhof M, Treurniet KM, Rinkel LA, Bruggeman AE, Berkhemer OA, Wolff L, van Voorst H, Tolhuisen ML, Dippel DWJ, et al; MR CLEAN–NO IV Investigators. A randomized trial of intravenous alteplase before endovascular treatment for stroke. *N Engl J Med*. 2021;385:1833–1844. doi: 10.1056/NEJMoa2107727
5. Fischer U, Kaesmacher J, Strbian D, Eker O, Cognard C, Plattner PS, Butikofer L, Mordasini P, Deppeler S, Pereira VM, et al; SWIFT DIRECT Collaborators. Thrombectomy alone versus intravenous alteplase plus thrombectomy in patients with stroke: an open-label, blinded-outcome, randomised non-inferiority trial. *Lancet*. 2022;400:104–115. doi: 10.1016/S0140-6736(22)00537-2
6. Mitchell RJ, Yan B, Churilov L, Dowling RJ, Bush SJ, Bivard A, Huo XC, Wang G, Zhang SY, Ton MD, et al; DIRECT-SAFE Investigators. Endovascular thrombectomy versus standard bridging thrombolytic with endovascular thrombectomy within 4.5 h of stroke onset: an open-label, blinded-endpoint, randomised non-inferiority trial. *Lancet*. 2022;400:116–125. doi: 10.1016/S0140-6736(22)00564-5
7. Gerschenfeld G, Liegey JS, Laborne FX, Yger M, Lyon V, Checkouri T, Tricard-Dessagne B, Marnat G, Clarençon F, Chausson N, et al. Treatment times, functional outcome, and hemorrhage rates after switching to tenecteplase for stroke thrombolysis: insights from the TETRIS registry. *Eur Stroke J*. 2022;7:358–364. doi: 10.1177/2396987322113729
8. Zhu F, Gauberti M, Marnat G, Bourcier R, Kyheng M, Labreuche J, Sibon I, Dargazanli C, Arquizan C, Anxionnat R, et al; ETIS Registry Investigators. Time from I.V. Thrombolysis to thrombectomy and outcome in acute ischemic stroke. *Ann Neurol*. 2021;89:511–519. doi: 10.1002/ana.25978
9. Tong X, Jia B, Ma G, Zhang X, Fiehler J, Flotmann F, Bechstein M, Brooks G, Hanning U, Kniep HC, et al; as the ANGEL-ACT & GSR-ET Study Group. Relationship between thrombolysis-to-puncture time and outcomes of endovascular thrombectomy in acute ischemic stroke. *Neurol Clin Pract*. 2025;15:e200434. doi: 10.1212/CPJ.0000000000000434
10. Tanswell P, Modi N, Combs D, Danays T. Pharmacokinetics and pharmacodynamics of tenecteplase in fibrinolytic therapy of acute myocardial infarction. *Clin Pharmacokinet*. 2002;41:1229–1245. doi: 10.2165/00003088-200241150-00001
11. Menon BK, Buck BH, Singh N, Deschaintre Y, Almekhlafi MA, Coutts SB, Thirunavukkarasu S, Khosravani H, Appireddy R, Moreau F, et al; ACT Trial Investigators. Intravenous tenecteplase compared with alteplase for acute ischaemic stroke in Canada (ACT): a pragmatic, multicentre, open-label, registry-linked, randomised, controlled, non-inferiority trial. *Lancet*. 2022;400:161–169. doi: 10.1016/S0140-6736(22)01054-6
12. Wang YJ, Li SY, Pan YS, Li H, Parsons MW, Campbell BCV, Schwamm LH, Fisher M, Che FY, Dai HG, et al; TRACE-2 Investigators. Tenecteplase versus alteplase in acute ischaemic cerebrovascular events (TRACE-2): a phase 3, multicentre, open-label, randomised controlled, non-inferiority trial. *Lancet*. 2023;401:645–654. doi: 10.1016/S0140-6736(22)02600-9
13. Parsons MW, Yogendrakumar V, Churilov L, Garcia-Esperon C, Campbell BCV, Russell ML, Sharma G, Chen C, Lin L, Chew BL, et al; TASTE Investigators. Tenecteplase versus alteplase for thrombolysis in patients selected by use of perfusion imaging within 4.5 h of onset of ischaemic stroke (TASTE): a multicentre, randomised, controlled, phase 3 non-inferiority trial. *Lancet Neurol*. 2024;23:775–786. doi: 10.1016/S1474-4422(24)00206-0
14. Meng X, Li S, Dai H, Lu G, Wang W, Che F, Geng Y, Sun M, Li X, Li H, et al. Tenecteplase vs alteplase for patients with acute ischemic stroke: the ORIGINAL randomized clinical trial. *JAMA*. 2024;332:1437–1445. doi: 10.1001/jama.2024.14721
15. Campbell BCV, Mitchell RJ, Churilov L, Yassi N, Kleinig TJ, Dowling RJ, Yan B, Bush SJ, Dewey HM, Thijs V, et al; EXTEND-IA TNK Investigators. Tenecteplase versus alteplase before thrombectomy for ischemic stroke. *N Engl J Med*. 2018;378:1573–1582. doi: 10.1056/NEJMoa1716405
16. Yogendrakumar V, Beharry J, Churilov L, Alidin K, Ugalde M, Pesavento L, Weir L, Mitchell RJ, Kleinig TJ, Yassi N, et al; Royal Melbourne Stroke Registry and EXTEND-IA TNK Collaborators. Tenecteplase improves reperfusion across time in large vessel stroke. *Ann Neurol*. 2023;93:489–499. doi: 10.1002/ana.26547
17. Wang L, Li J, Wu X, Li L, Jiao X, Che F, Han H, Zhang L, Liu W, Wang P, et al. Tenecteplase compared to alteplase before mechanical thrombectomy enhances 1-h recanalization and reduces disability in large-vessel occlusion. *J Neurol*. 2025;272:324. doi: 10.1007/s00415-025-13084-2
18. Tang F, Langenhorst J, Dang S, Kassir N, Owen R, Purdon B, Magnusson MOO, Deng R. Population pharmacokinetics of tenecteplase in patients with acute myocardial infarction and application to patients with acute ischemic stroke. *J Clin Pharmacol*. 2023;63:197–209. doi: 10.1002/jcph.2164
19. Kobayashi Y, Saito T, Kawabata Y, Igari T, Kondo Y, Yamamoto K, Sekijima Y. A 5-minute delay from bolus injection to infusion halves plasma concentration of tissue plasminogen activator. *J Emerg Med*. 2023;64:709–713. doi: 10.1016/j.jemermed.2023.03.059



20. Smith C, Al-Nuaimi Y, Wainwright J, Sherrington C, Singh A, Kallingal J, Douglass C, Parry-Jones A, Smith C, Dixit A, et al. The influence of bolus to infusion delays on plasma Tissue Plasminogen Activator levels. *Int J Stroke*. 2014;9:939–942. doi: 10.1111/j.1747-4949.2012.00924.x
21. Hacke W, Kaste M, Fieschi C, von Kummer R, Davalos A, Meier D, Larrue V, Bluhmki E, Davis S, Donnan G, et al. Randomised double-blind placebo-controlled trial of thrombolytic therapy with intravenous alteplase in acute ischaemic stroke (ECASS II). Second European-Australasian Acute Stroke Study Investigators. *Lancet*. 1998;352:1245–1251. doi: 10.1016/s0140-6736(98)08020-9
22. Mazya M. Predicting the risk of symptomatic intracerebral hemorrhage in ischemic stroke treated with intravenous alteplase: Safe Implementation of Treatments in Stroke (SITS) symptomatic intracerebral hemorrhage risk score (vol 43, pg 1524, 2012). *Stroke*. 2012;43:E102–E102. doi: 10.1161/STR.0b013e318262de57
23. Gerschenfeld G, Turc G, Obadia M, Chausson N, Consoli A, Olindo S, Caroff J, Marnat G, Blanc R, Ben Hassen W, et al; for ETIS and TETRIS study groups. Functional outcome and hemorrhage rates after bridging therapy with tenecteplase or alteplase in patients with large ischemic core. *Neurology*. 2024;103:e209398. doi: 10.1212/WNL.00000000000209398
24. Checkouri T, Gerschenfeld G, Seners P, Yger M, Ben Hassen W, Chausson N, Olindo S, Caroff J, Marnat G, Clarencon F, et al; TETRIS and PREDICT-RECANAL Collaborators. Early recanalization among patients undergoing bridging therapy with tenecteplase or alteplase. *Stroke*. 2023;54:2491–2499. doi: 10.1161/STROKEAHA.123.042691
25. Seners P, Caroff J, Chausson N, Turc G, Denier C, Pottin M, Aghasaryan M, Alecu C, Chassin O, Lapergue B, et al; PREDICT-RECANAL Collaborators. Recanalization before thrombectomy in tenecteplase vs. alteplase-treated drip-and-ship patients. *J Stroke*. 2019;21:105–107. doi: 10.5853/jos.2018.01998
26. Yogendrakumar V, Churilov L, Guha P, Beharry J, Mitchell PJ, Kleinig TJ, Yassi N, Thijs V, Wu TY, Brown H, et al; EXTEND-IA TNK Investigators. Tenecteplase treatment and thrombus characteristics associated with early reperfusion: an EXTEND-IA TNK trials analysis. *Stroke*. 2023;54:706–714. doi: 10.1161/STROKEAHA.122.041061
27. Goyal M, Menon BK, van Zwam WH, Dippel DWJ, Mitchell PJ, Demchuk AM, Davalos A, Majoie CBLM, van der Lugt A, de Miquel MA, et al; HERMES Collaborators. Endovascular thrombectomy after large-vessel ischaemic stroke: a meta-analysis of individual patient data from five randomised trials. *Lancet*. 2016;387:1723–1731. doi: 10.1016/S0140-6736(16)00163-X
28. Lima FO, Furie KL, Silva GS, Lev MH, Camargo EC, Singhal AB, Harris GJ, Halpern EF, Koroshetz WJ, Smith WS, et al. The pattern of leptomeningeal collaterals on CT angiography is a strong predictor of long-term functional outcome in stroke patients with large vessel intracranial occlusion. *Stroke*. 2010;41:2316–2322. doi: 10.1161/STROKEAHA.110.592303
29. Maas MB, Lev MH, Ay H, Singhal AB, Greer DM, Smith WS, Harris GJ, Halpern E, Kemmling A, Koroshetz WJ, et al. Collateral vessels on CT angiography predict outcome in acute ischemic stroke. *Stroke*. 2009;40:3001–3005. doi: 10.1161/STROKEAHA.109.552513
30. Paoni NF, Keyt BA, Refino CJ, Chow AM, Nguyen HV, Berleau LT, Badillo J, Peña LC, Brady K, Wurm FM. A slow clearing, fibrin-specific, PAI-1 resistant variant of t-PA (T103N, KHRR 296-299 AAAA). *Thromb Haemost*. 1993;70:307–312.
31. Benedict CR, Refino CJ, Keyt BA, Pakala R, Paoni NF, Thomas GR, Bennett WF. New variant of human Tissue Plasminogen Activator (TPA) with enhanced efficacy and lower incidence of bleeding compared with recombinant human TPA. *Circulation*. 1995;92:3032–3040. doi: 10.1161/01.cir.92.10.3032
32. Bhatia R, Hill MD, Shobha N, Menon B, Bal S, Kochar P, Watson T, Goyal M, Demchuk AM. Low rates of acute recanalization with intravenous recombinant tissue plasminogen activator in ischemic stroke: real-world experience and a call for action. *Stroke*. 2010;41:2254–2258. doi: 10.1161/STROKEAHA.110.592535
33. Ahn SH, Hong R, Choo IS, Heo JH, Nam HS, Kang HG, Kim HW, Kim JH. Histologic features of acute thrombi retrieved from stroke patients during mechanical reperfusion therapy. *Int J Stroke*. 2016;11:1036–1044. doi: 10.1177/1747493016641965
34. Meretoja A, Keshtkaran M, Tatlisumak T, Donnan GA, Churilov L. Endovascular therapy for ischemic stroke: save a minute-save a week. *Neurology*. 2017;88:2123–2127. doi: 10.1212/WNL.0000000000003981
35. Cui C, Qiu J, Zhao YG, Chen HS. Thrombolysis-to-recanalization time and clinical outcomes in large-vessel occlusion patients with successful recanalization. *J Neurol*. 2025;272:687. doi: 10.1007/s00415-025-13440-2
36. Fassbender K, Dempfle CE, Mielke O, Schwartz A, Daffertshofer M, Eschenfelder C, Dollman M, Hennerici M. Changes in coagulation and fibrinolysis markers in acute ischemic stroke treated with recombinant tissue plasminogen activator. *Stroke*. 1999;30:2101–2104.
37. Rha J-H, Saver JL. The impact of recanalization on ischemic stroke outcome: a meta-analysis. *Stroke*. 2007;38:967–973. doi: 10.1161/01.STR.0000258112.14918.24
38. Arrarte Terreros N, Bruggeman AAE, Swijnenburg ISJ, van Meenen LCC, Groot AE, Coutinho JM, Roos YBWEM, Emmer BJ, Beenen LFM, van Bavel E, et al. Early recanalization in large-vessel occlusion stroke patients transferred for endovascular treatment. *J Neurointerv Surg*. 2022;14:480. doi: 10.1136/neurintsurg-2021-017441
39. Yang P, Zhang Y, Zhang L, Zhang Y, Treurniet KM, Chen W, Peng Y, Han H, Wang J, Wang S, et al; DIRECT-MT Investigators. Endovascular thrombectomy with or without intravenous alteplase in acute stroke. *N Engl J Med*. 2020;382:1981–1993. doi: 10.1056/NEJMoa2001123
40. Qiu Z, Li F, Sang H, Yuan G, Xie D, Zhou K, Li M, Meng Z, Kong Z, Ruan Y, et al; BRIDGE-TNK Trial Investigators. Intravenous tenecteplase before thrombectomy in stroke. *N Engl J Med*. 2025;393:139–150. doi: 10.1056/NEJMoa2503867